

Elicitation Techniques

Chapter 2: Elicitation Skills & Techniques

Chapter Objectives

In this chapter, we will explore the elicitation skills required to be able to effectively apply elicitation techniques by:

- Examining some different techniques that can be used for elicitation
- Identifying skills such as listening, body language, etc.
- Evaluating derived requirements with respect to structure and content

Elicitation Skills

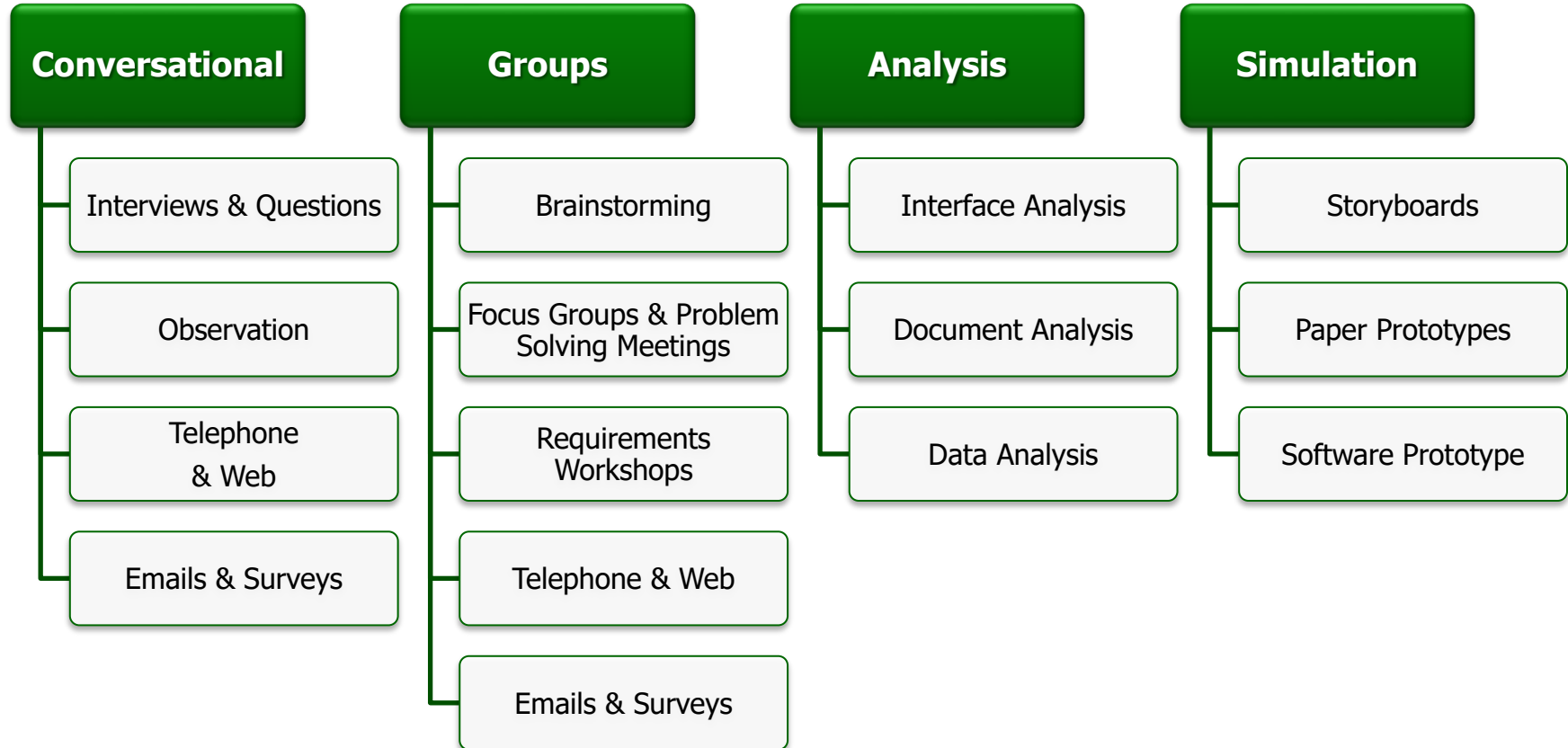


Techniques for Elicitation

Human Skills for Elicitation

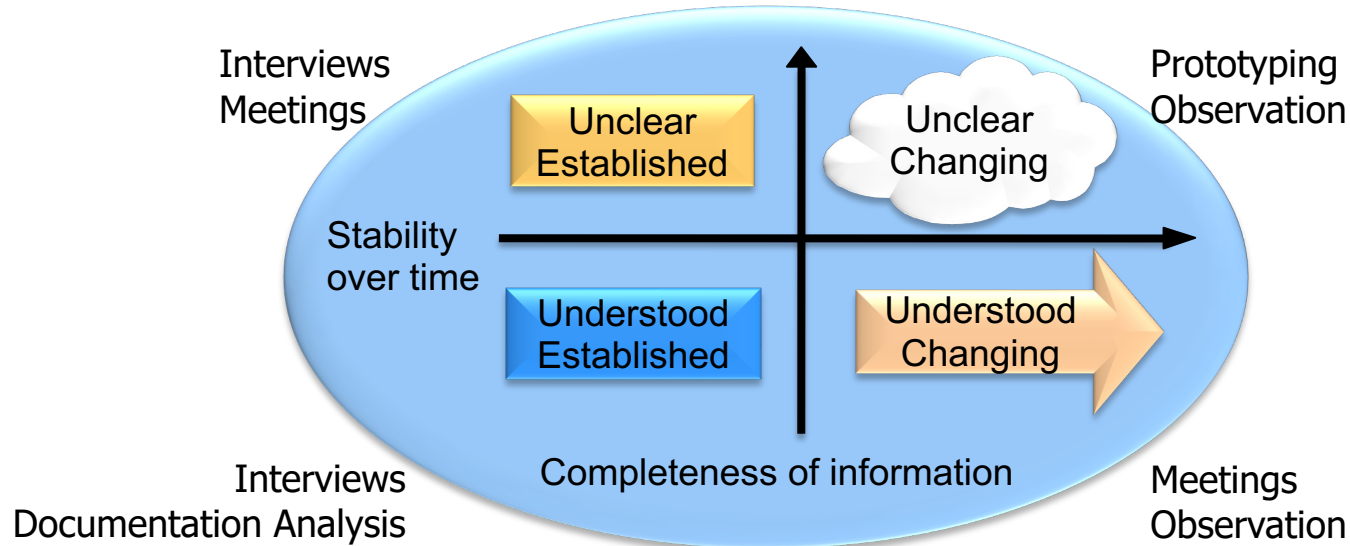
Evaluating Requirements

Elicitation Techniques



Choosing an Elicitation Approach

- Assess the stability and completeness of information sources
- Choose an information gathering approach that matches



Interface Analysis

- A technique that looks for the inputs, outputs, and connections between
 - Human to computer interfaces (HCI)
 - Systems to system interfaces
 - Interfaces to external hardware devices
- Interface analysis answers the “what” and “why” of interfaces
 - Interface design addresses the “how”
- Analysis may include identifying all the fields in an existing GUI
 - And the backend source of the data
- Analysis may include identifying all the data used in particular report



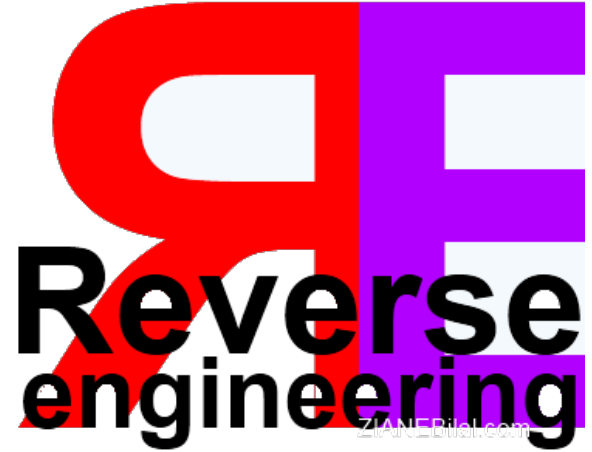
Document Analysis



- Document analysis makes use of all the document-based artifacts available on a project
 - Also called document research
- The reading and understanding of the case study manual for this program is an example of performing a document analysis
- Reading the “existing stuff” can push your project knowledge forward very quickly and without needing interaction from project team members
- For QA – document analysis is a key part of the ‘verify and validate’ as the starting points are the requirements for verification and the specifications for validation

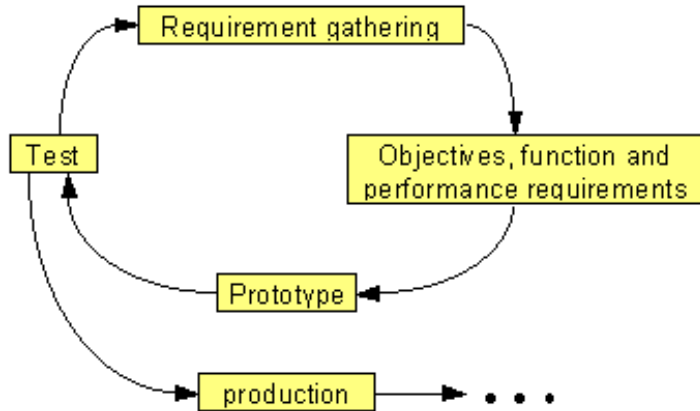
Reverse Engineering

- Reverse engineering is the process of analyzing a subject system to create representations of the system at a higher level of abstraction*
- Use code to develop designs on which the code is based
 - Reverse engineer the designs and code back to requirements
- Also applied to creating requirements from operational system
 - Documenting the interface requirements by executing the system



*E. J. Chikofsky and J. H. Cross, II, "Reverse Engineering and Design Recovery: A Taxonomy," *IEEE Software*, vol. 7, January 1990, no. 1, pp. 13-17

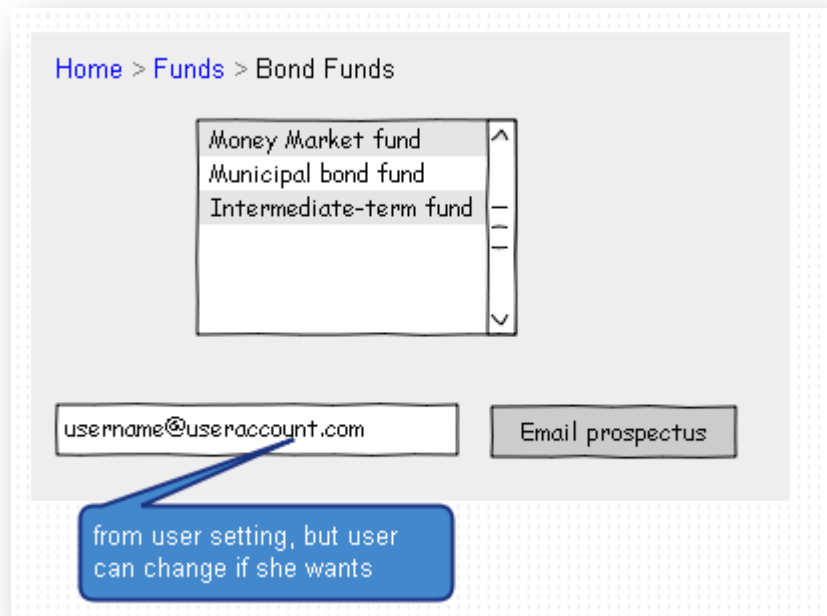
Prototyping



- A prototype is a preliminary type, form, or instance of a product or product component
 - Can serve as a model for later stages
 - Can evolve into the final, complete version of the product
- Prototypes are used for the following:
 - Assessing the feasibility of a new or unfamiliar technology
 - Assessing or mitigating technical risk
 - Validating requirements
 - Demonstrating critical features
 - Qualifying a product
 - Qualifying a process
 - Characterizing performance or product features

Wireframe

- A wireframe is a schematic representation of a UI
 - Uses sketches of elements
 - Allows for planning before coding
 - Similar to a prototype
 - Wireframes contain no functionality
 - Prototypes sometimes contain functionality
- Wireframes are useful to capture ideas
 - Useful in brainstorming when walking through possibilities
 - Allow communication between different stakeholders
 - Best if a tool is used to capture final plan



Elicitation Skills

Techniques for Elicitation



Human Skills for Elicitation

Evaluating Requirements

The Skills Behind the Techniques

- The human-to-human technique of gathering information is often the hardest for people to learn
 - Conversations
 - Interviewing
 - Facilitating meetings
- To be successful at elicitation, it takes more than just applying the knowledge of a set of techniques
 - It takes an often subtle skill set to be an effective elicitor



Miller's Law

In order to understand what another person is saying, you must assume that it is true and try to find out what it could be true of.

—George Miller

- Not accepting that the statement is true
- You're assuming temporarily that it is true, and then you're listening carefully to find out what the statement could be true *of*
- Give the speaker your full attention for as long as it takes to understand what's really being said and why
- Seek to understand someone's view rather than making interpretations

The Listener LADDER

- **L**ook at the person speaking to you
- **A**sk questions
- **D**on't interrupt
- **D**on't change the subject
- **E**mpathize
- **R**espond verbally and non-verbally



What Is Nonverbal Communication?



- The majority of the messages people communicate are nonverbal
- Nonverbal communication is also known as body language
- Includes
 - Facial expressions
 - Gestures
 - Eye contact
 - Posture
 - Tone of our voice
 - Use of space
 - Body movements
 - Touch

Five Roles of Nonverbal Communication

- Repetition
 - Repeats the message the person is making verbally
- Contradiction
 - Contradicts the message being conveyed verbally
- Substitution
 - Substitutes for the verbal message
- Complements
 - Adds value or aides a verbal message
- Accents
 - Punctuates a verbal message



The Importance of Effective Communication, Edward G. Wertheim, Ph.D

Body Language

- Posture
- Space taken up by individual
- Hand-to-face movements
 - Typically indicates verge of decision making
- Gestures
 - Head moves to gain attention
 - Display of underbelly shows confidence, security or trust
 - Grooming gestures are done for reassurance
 - Hands
 - Pyramid shape means power



- Displacement activity
 - Used to get rid of physical tension in the body
 - Nail biting, chewing gum, teeth grinding

- Eye contact
 - Assumed to be indicator of truthfulness
 - Culturally impacted



Optional Activity 2.1: Active Listening



Goal	To paraphrase a statement during an elicitation meeting
Team Format	Teams of three
Materials	Next page
Tasks	1. Follow instructions on next page 2. After completion change roles and do it again 3. Exercise is complete when all participants have performed all roles
Deliverable	Summary of observer notes to be reported to class as whole
Duration	45 minutes

Activity 2.1: Active Listening Task List



- 1. Each participant is assigned one role:
 - Player, responder, observer
- 2. Player: Think of a time when you were emotionally charged (happy, mad, etc.)
 - Consider who you might have liked to share it with
 - Tell the responder what role they will play, but not what the situation is about
- 3. Player: Imagine yourself in the situation, and about to meet the responder
 - Place yourself in that mindset
- 4. Responder: Actively listen to the player
 - Defuse the emotions, paraphrase, reach consensus
- 5. Observer: Prepare to critique Responder's active listening techniques
- 6. Time limit: 10 mins

Meetings

- Meetings are the ultimate form of managed conversation
- Meetings come in a variety of different flavors
- Statistics show that managers spend about 50% of their day in meetings
 - For senior managers it may be as much as $\frac{3}{4}$ of a day
- Meetings are not a substitute for one-on-one interchange
 - Fallacy to think that it can speed up the process



The Pros and Cons of Meetings



■ Pros

- Gather more information faster
- Resolve conflict
- Gain confirmation more efficiently and completely

■ Cons

- Depends on facilitation
- May be subject to “group think”
- Some with information may not volunteer in group setting
- One member of meeting may dominate
 - Or one political position

■ Use meetings to gather or confirm information

- Not to define requirements
- Always start the session with the problem and vision

Planning a Meeting

- Choose the type of meeting
 - Live or virtual
 - Plan the location of live meetings to ensure
 - Meeting room size is adequate
 - Meeting room equipment is adequate
- Decide the objectives of session
 - Stay focused
 - Ensure multiple objectives are related
- Determine participants and their roles
 - Invite participants and limit to those who can help with objectives
 - Distinguish must attend vs. can attend
- Create a prioritized agenda
 - Publish prior to meeting if appropriate
 - Use in the meeting to set the ground rules



Hosting the Meeting

- Set the professional tone for the meeting by example
 - Start and stop the meeting on time
 - Establish and enforce the meeting rules
 - Introduce goals based on the agenda
- Keep the meeting on track
 - At a divergence from meeting objectives, re-focus on original goals
 - Schedule a different meeting to deal with tangential topics
 - I'd like to take that topic off-line
 - Let's you and I meeting individually to dig deeper into that topic
- Facilitate decision making and consensus making
 - Avoid participating in the decision or consensus
 - Ensure all stakeholders participate
 - Ensure all stakeholders have their input heard
 - Actively manage disruptive behaviors

Requirement Breakdown Workshop

- A structured interviewing meeting with a small group of stakeholders
- Objectives
 - Discover and elicit new requirements
 - Decompose requirements into smaller entities
- When is it appropriate?
 - When requirements are not well understood
 - When new technology is being deployed
 - High risk projects
 - Generally used in Waterfall
- Participants include
 - Analyst and relevant stakeholders (SME, Users, etc.)



Optional Activity 2.2: Requirement Breakdown Workshop



Goal	Build an RBS using a Requirement Breakdown Workshop
Team Format	Teams of 5 or 6
Materials	Flip chart and information from the next slide
Tasks	1. Your team has been selected to detail the user requirements for the LEAP Social Media web site (see next page) 2. Using a functional decomposition approach, gather as many meaningful requirements as possible 3. Create a graphic of your RBS 4. Prepare to present your material
Deliverable	A Requirement Breakdown Structure
Duration	20 minutes (create the document), 2 minutes each to present the plan to the class

Activity 2.2: Requirement Breakdown Meeting



■ Leap Social Networking Site

- Our idea is to build a site that will encourage past, present, and future Leap associates to keep in touch, build relationships, and share information in a way that will make your time at company more effective, but also more enjoyable. But we need your help. What are the sort of services would such a site have to offer to make it something that you would want to use on a regular basis?

Elicitation Skills

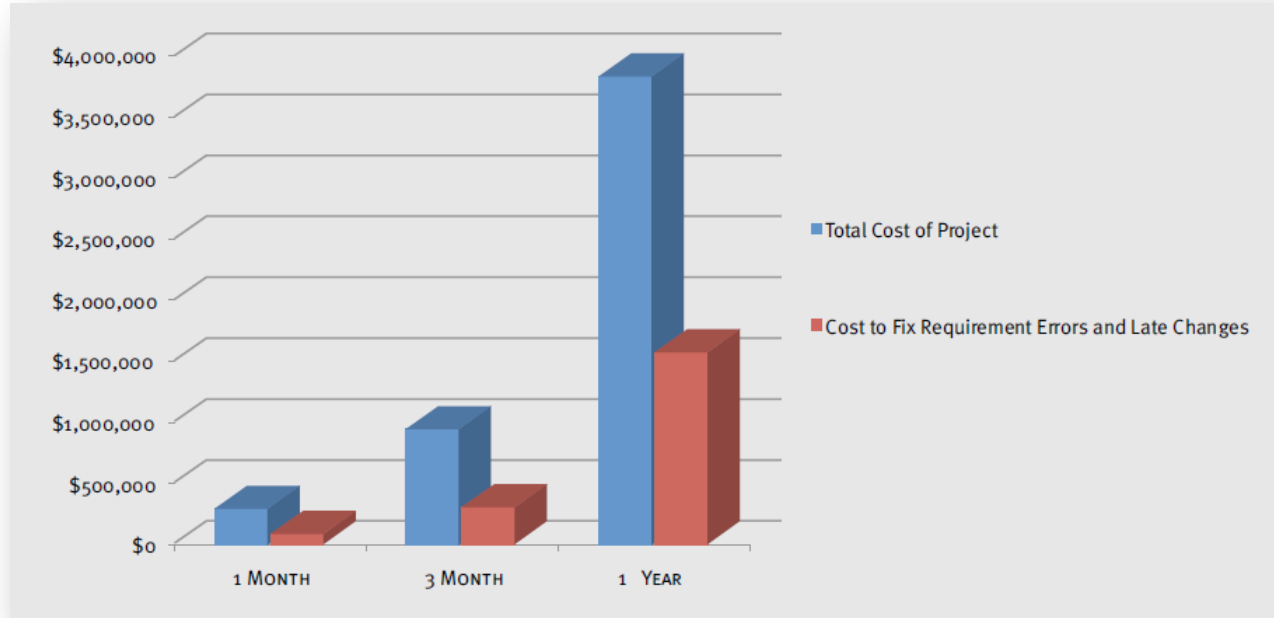
Techniques for Elicitation

Human Skills for Elicitation



Evaluating Requirements

Industry Insight: Cost of Poor Requirements

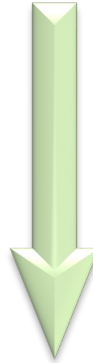


Cost of neglecting requirements on a 25 person project

Borland, Bringing Software Requirements to life, White Paper 2009

Levels of Evaluation

- Peer review
 - Manager and analysts
 - Document-oriented
 - Focus is improving quality of requirements
- Requirements Walkthrough
 - Analysts and stakeholders
 - Document-oriented
 - Focus on missed requirements



- Requirements Demonstration
 - Analysts and stakeholders
 - Prototyping and storyboards
 - Focus on missed requirements
- Audit/Inspection
 - External inspector
 - Focus on requirements and process

**More complete
of defects
More rigorous
types of defects**

Requirements Evaluation—Are they SMART and Testable?

■ SMART

- **S**: Specific
- **M**: Measurable
- **A**: Attainable
- **R**: Realistic
- **T**: Time bound

■ Testable: A testable requirement has

been broken down to be:

- Precise
- Unambiguous
- Not divisible into lower-level requirements

How to confirm:

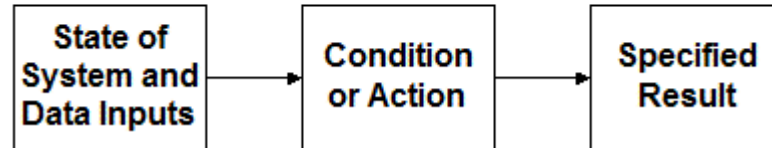
- Can a test case be written to validate the the requirement has been implemented correctly?

—Definition from: [Mosaic Inc.](#)

Testable Requirements—*Reference Material*

- For most requirements, a testable requirement could be described in terms of:
 - The state of the system and the data elements that are inputs
 - For example, customer number, product number
 - The condition or action associated with the requirement
 - For example, the user enters data, the order is validated, the check amount is deducted
 - The expected or specified result described in terms of data elements
 - For example, customer number must be 8-digit numeric, quantity must be greater than zero

Testable Requirements



Standard of Quality

■ The standard of a high quality requirement is that each requirement is:

- Complete
- Consistent
- Unambiguous
- Well structured
- Correct
- Feasible
- Verifiable
- Traceable
- Necessary
- Prioritized
- Modifiable



* Adapted from IEEE Std 830-1998

Complete

- A set of requirements is considered complete if it addresses all significant areas of functionality required to solve the original business problem
- Ensure that requirements are complete
 - Compare the set of requirements to the project's problem/opportunity and agreed upon scope
 - If the requirements are implemented, will they completely solve the problem or allow the opportunity to be seized?
- All references to the phrase "To Be Determined (TBD)" have been resolved
 - Specifics of owner and due date can be used
- Completeness often cannot be determined without stakeholder review

Consistent



- Each requirement must be in agreement with, or at least not in contradiction with, another requirement
- When two requirements conflict
 - Through negotiation with the user community, change one requirement to agree with the other
 - Modify one requirement with additional information to detail special
 - Prioritize the requirements so that one will be applied before the other(s)
- Use the same wording throughout for actors and system names

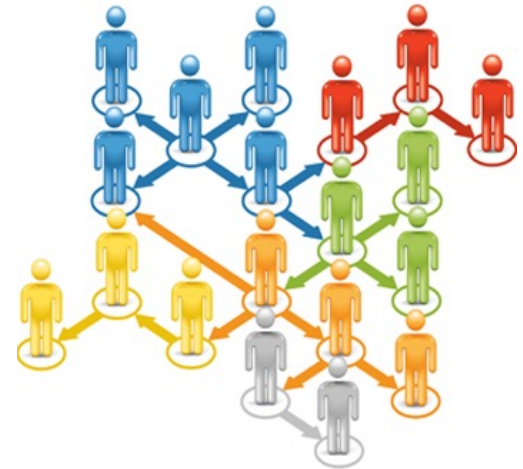
Unambiguous

- If a requirement can be interpreted in more than one way, there is ambiguity
 - Needs to be rewritten clearly
- Each requirement should address exactly one capability
- Requirements should be understood by a 12-year old
 - 7th or 8th grade English skills
- Word choice matters
- To be discussed in greater detail in next section



Well Structured

- Good format to follow: <Optional Condition><Subject>,<focus word>, <capability>, criteria
 - Condition: Used as an option to define when or where the requirement will be used
 - Subject: Who is doing something
 - Focus word: type of requirement (shall, may, must, etc.)
 - Capability: what is to be done
 - Criteria: conditions or constraints
 - To be covered in more detail later in this chapter
- Written in an Active voice, not Passive
- Good numbering system
- Proper spelling and grammar
- To be discussed further, later in this chapter

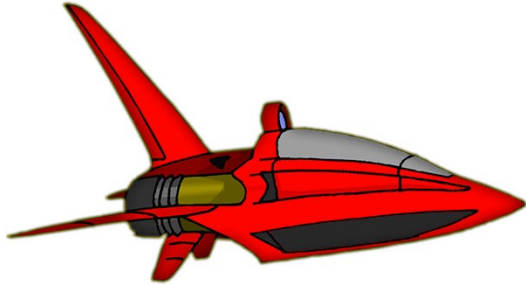




Correct

- All business calculations are error free
 - Determined through frequent review
 - Provided by a legitimate authority
- Individual requirements have been inspected for conflict
 - All conflicts resolved with appropriate stakeholders
- Verify all requirements with second source
 - Confirm through documentation
 - Observe with additional stakeholders

Feasible and Verifiable



■ Feasible

- It must be possible to satisfy the requirement
 - Within the limitations of the environment
 - With all possible software engineering capabilities

■ Verifiable

- It must be possible to verify that the requirement has been satisfied
 - With a test with known input and expected output
 - Watch for “always” and “never”
 - Some other means
 - Inspection
 - Analysis
 - Demonstration

1 + 1 = _

Traceable

- Trace requirements to initial source
 - Use Case
 - Policy, procedure, business rule
- Requirements derive from a problem statement
- A requirements document must only address those requirements necessary to solve the problem

Policy Information will be available



12.4 The policy information will be accessed
12.4.1 by entering policy number
12.4.2 by entering customer ID

Prioritized—in Rank Order and Necessary



- Identify the importance or stability of the requirement
 - Make sure all requirements are **necessary**
 - Required by the customer
 - Required to be in conformance
- Apply a consistent **prioritization** to requirements
- If there is doubt about the necessity of the requirement, ask
 - What is the worst thing that could happen if this requirement were not included?
 - Ask if removing a requirement will create a deficiency
- All requirements should be part of a solution to the problem

Modifiable

- Every requirement should have a unique identifier
 - Number the requirements
 - One requirement per statement
- Use a hierarchy to group related requirements
 - UR 1.5
 - UR 1.5.1
 - UR 1.5.2
 - UR 1.5.2.1
 - UR 1.5.2.2
 - ~~– UR 1.5.2.3~~
 - UR 1.5.2.4
- If a requirement is removed from scope, strike through the unique identifier and do not reuse it



Activity 2.3: Write An Excellent Requirement



Goal	Write a requirement that communicates well
Team Format	Teams of 4 from previous activity
Materials	This slide, flip chart from previous activity
Tasks	1. Consider your RBS, from the last activity 2. Write five “excellent” requirements for the project 3. Define your requirements as BR, UR, SR 4. Evaluate your requirements with respect to their Standard of Quality 3. Be prepared to defend your standards
Deliverable	Five written requirements on a flip chart
Duration	30 minutes

Technique: Verification Checklist

Attribute	Explanation
Complete	No requirements or necessary information should be missing
Consistent	Consistent requirements do not conflict with other requirements
Unambiguous	The reader of a requirement draw one interpretation, and satisfies a single requirement
Well structured	Each requirement has subject, condition, capability, criteria
Correct	Each requirement must accurately describe functionality to be delivered
Feasible	Each requirement can be implemented within the constraints of the project
Verifiable	Confirm you can devise tests or use inspection, analysis, or demonstration to see the requirement is satisfied
Traceable	Ability to link each requirement to its source or other related requirements
Necessary	Each requirement should document something the customers really need or to be in conformance
Prioritized	Assign an implementation priority (rank) to each requirement or use case to indicate how important it will be
Modifiable	Assign a unique identifier allowing cross referencing rather than duplication

Chapter Summary

In this chapter, we have explored the elicitation skills required to be able to effectively apply elicitation techniques by:

- Examining some different techniques that can be used for elicitation
- Identifying skills such as listening, body language, etc.
- Evaluating derived requirements with respect to structure and content – to be SMART and testable